Concordia University

COMP 353 - X

Warm up Project Report

Names

[redacted]

[redacted]

Solomon Lewis (40257710)

[redacted]

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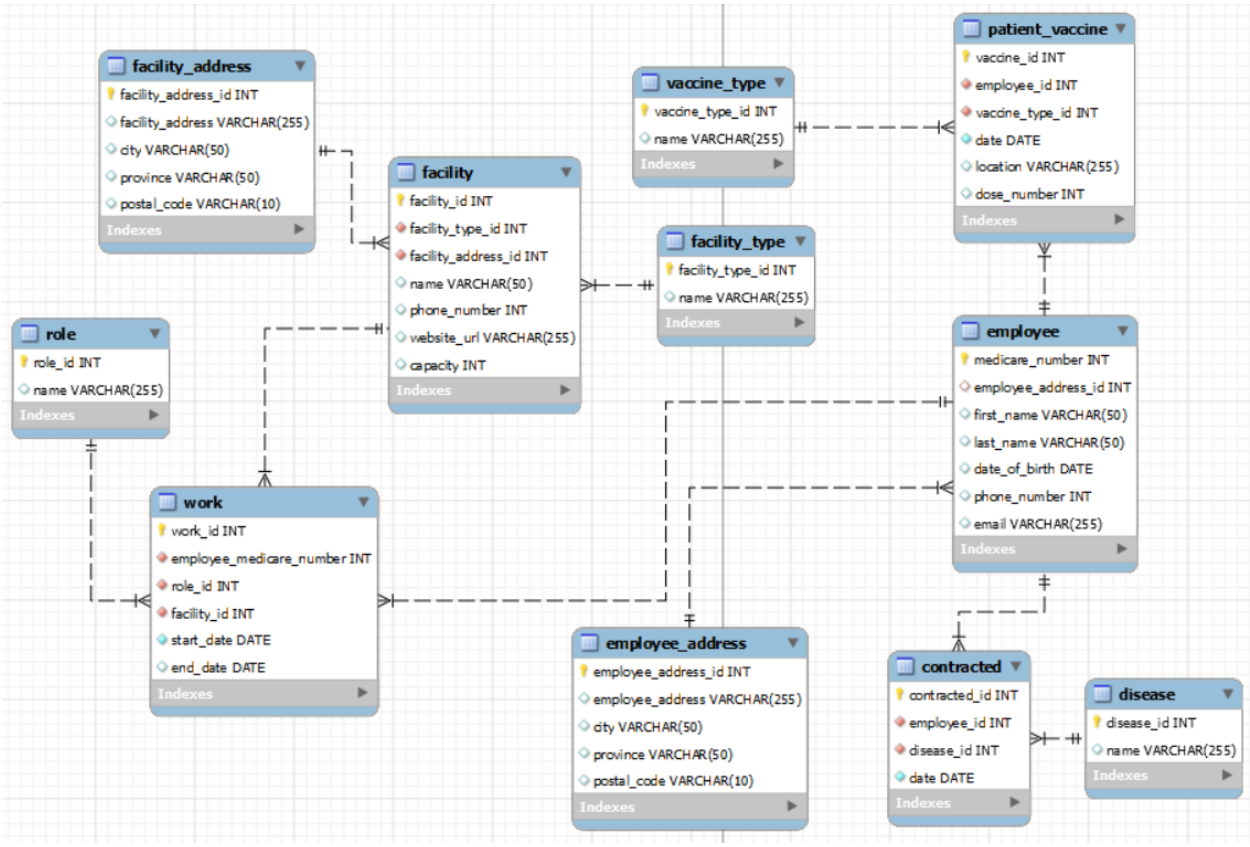
We certify that this submission is the original work of members of the group and meets the Faculty's Expectations of Originality

1. Design



Key:

- Yellow: Primary Key
- Red: Foreign Key

Facility Table

The ‘facility’ table contains data to represent a particular healthcare facility. The facility_type is then linked by a relation to the ‘facility’ table by use of a foreign key. The ‘facility_address’ table is linked to the ‘facility’ table and used to represent the address information of the healthcare facility. The ‘facility_address_id’ attribute is used to keep track of the particular address attached to the ‘facility’ and is stored as a foreign key of the table.

Facility_Adress Table

The 'facility_address' table contains the data to represent the address of each facility. The table is linked to the 'facility' table using the foreign key 'facility_address_id'.

Facility_Type Table

The 'facility_type' table contains the names of all the healthcare facilities that exist. This could include types such as hospital, clinic, pharmacy. The primary key of this table is then linked to the facility table using the tables primary key ‘facility_type_id’.

Work Table

The ‘work’ table contains data to represent the details of the employees working at a particular healthcare facility. To link each employee to a particular job, the primary key of each employee (‘employee_medicare_number’) is included as a foreign key attribute. The role of each employee is tracked through the role_id attribute foreign key from the ‘role’ table. The table also tracks which facility each employee is working in using the foreign key from the ‘facility’ table. Attributes that track the ‘start_date’ and ‘end_date’ of each employee’s work are included.

Role Table

The ‘role’ table contains a ‘name’ used to track all the different roles an employee can have. This could include, doctor, nurse, receptionist, or any other job titles that may be added in the future. Each row contains a tuple where the ‘role_id’ attribute links to the role name.

Employee Table

The ‘employee’ table contains data to represent each employee working at some healthcare facility. The primary key of the table ‘employee_medicare_number’ is used to uniquely identify each employee. The table includes attributes to represent the requested data in the requirements attached to each employee, such as their name, date of birth, phone number and email.

Employee_Address Table

The ‘employee_address’ table contains data to represent the address of each employee. The primary key of the table is linked to the ‘employee’ table as a foreign key, represented in the employee table as ‘employee_address_id’. The attributes of this table are used to represent the actual data for the address of each employee.

Patient_Vaccine Table

The ‘patient_vaccine’ table contains data to represent a vaccine an employee has received. The table keeps track of the type of vaccine, the date of the vaccination, the dose number and the id of the employee who has taken the vaccine. The ‘employee_id’ attribute is stored as a foreign key and tracks the primary key ‘employee_medicare_number’ from the ‘employee’ table to track each individual employee.

Vaccine_Type Table

The ‘vaccine_type’ table is used to represent the names of all vaccines an employee can have, such as Pfizer, AstraZeneca, etc. The primary key of this table ‘vaccine_type_id’ is linked to the ‘patient_vaccine’ table as a foreign key and stored to keep track of the particular vaccine each employee has taken. Additional types may be added to this table in the future and may be converted into additional sub-tables to represent the different categories of vaccines an employee may be able to receive.

Contracted Table

The ‘contracted’ table contains the details of a particular disease that may be contracted by an employee. The table keeps track of the name of the disease contracted and the date of contraction. The ‘employee_id’ foreign key attribute keeps track of which employee contracted a particular disease and the ‘disease_id’ foreign key attribute tracks the type of disease contracted.

Disease Table

The ‘disease’ table contains the different types of known diseases an employee can contract, such as COVID-19, SARS-Cov-2, etc. The primary key of this table is linked to the ‘contracted’ table and is stored in the ‘disease_id’ attribute of the table.

Summary of the Sub-Table Choices

To improve the normalization of the database and avoid data redundancy we added sub-tables to the database. We did this to organize the data in a way that reduces data redundancy and dependencies, ultimately improving data integrity and maintaining consistency within our database. Splitting up the tables into sub-tables helps to avoid repeated data that could span multiple rows, minimizing data redundancy.

For example the ‘employee’ and ‘facility’ tables could have included the address as additional attributes contained in each schema, but instead of potentially repeating the data we created separate sub-tables which are linked to the tables using foreign keys.
Similarly the ‘work’, ‘vaccine’ and ‘disease’ tables could have contained additional attributes to represent their type, but splitting this into sub-tables improves the accuracy of the data and ensures that no data is repeated within the database.

2. SQL CREATE statements

Employee Table

```
/* EMPLOYEE TABLE */
CREATE TABLE employee (
    medicare_number INT NOT NULL AUTO_INCREMENT,
    employee_address_id INT NOT NULL,
    first_name varchar(50) NOT NULL,
    last_name varchar(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    phone_number INT,
    email varchar(255),

    FOREIGN KEY (employee_address_id) REFERENCES employee_address(employee_address_id),
    PRIMARY KEY (medicare_number)
);
```

The medicare number is the primary key of this table. Each employee has their own unique medicare number, and it is used to differentiate them in many tables. It was decided that the medicare number cannot be NULL as every employee must have a unique registered medicare number in order to work at their respective facility. This table relates to the employee's address through the foreign key 'employee_address_id', which is the primary key of its own table. The employee_address_id cannot be NULL as if an employee exists they must have an address where they live. This table contains the personal attributes of each employee, namely their first/last names, their date of birth, their phone number, and their email. The first and last names were decided to be of type VARCHAR with a length of 50 to accommodate employees with large names. Date of birth is of type DATE, phone number is of type INT and email is of variable type VARCHAR with a length of 255 to accommodate those with large emails. It is possible that these values can be NULL as an employee may not have a phone number or an email address.

Employee_Address Table

```
/* EMPLOYEE ADDRESS TABLE */
CREATE TABLE employee_address (
    employee_address_id INT NOT NULL AUTO_INCREMENT,
    employee_address varchar(255) NOT NULL,
    city varchar(50) NOT NULL,
    province varchar(50) NOT NULL,
    postal_code varchar(10) NOT NULL,

    PRIMARY KEY (employee_address_id)
);
```

The employee address ID is the primary key for this table. It is of type INT and cannot be NULL since if an employee exists, they must have some sort of ID associated with their address in the database. The actual address itself is in the attribute ‘employee_address’ and it is of type VARCHAR with length 255 to accommodate for large addresses. The attribute city is of type VARCHAR with length 50, attribute province is of type VARCHAR with length 50, and postal code is of VARCHAR with length 10. These values cannot be NULL as an address is composed of a city, a province and a postal code.

Facility Table

```
/* FACILITY TABLE */
CREATE TABLE facility (
    facility_id INT NOT NULL AUTO_INCREMENT,
    facility_type_id INT NOT NULL,
    facility_address_id INT NOT NULL,
    name varchar(50) NOT NULL,
    phone_number INT,
    website_url varchar(255),
    capacity INT,

    FOREIGN KEY (facility_type_id) REFERENCES facility_type(facility_type_id),
    FOREIGN KEY (facility_address_id) REFERENCES facility_address(facility_address_id),
    PRIMARY KEY (facility_id)
);
```

The primary key for the facility table is ‘facility_id’ and it is of type INT. This attribute uniquely identifies what facility an employee works at. Facility_type_id is an attribute of type INT that is a foreign key; it relates to the table ‘facility_type’ and it distinguishes what kind of facility the employee works at. ‘Facility_address_id’ is also a foreign key attribute of type INT and it relates to the ‘facility_address’ table which holds the address information of the facility. The ‘name’ attribute holds the name of the facility and is of type VARCHAR with a length of 50. The attribute ‘phone_number’ holds the facility’s phone number and is of type INT. The phone number may be NULL as it is possible a facility does not have a phone number. The attribute ‘website_url’ is of type VARCHAR of length 255. This attribute holds the link to the facility’s website and can also be NULL as it is possible a facility does not have a website. Capacity is an attribute of type INT which indicates the maximum number of employees required to work at this facility. The capacity may be NULL as it is possible that there are no employees working at a facility.

Facility_type Table

```
CREATE TABLE facility_type (
    facility_type_id INT NOT NULL AUTO_INCREMENT,
    name varchar(255) NOT NULL,

    PRIMARY KEY (facility_type_id)
);
```

The primary key for this table is ‘facility_type_id’ and it is of type INT. This attribute uniquely identifies the type of facility it is (hospital, clinic, etc). The attribute ‘name’ is of type VARCHAR with a length 255 and cannot be NULL as facilities must be named a certain type.

Facility Address Table

```
/* FACILITY ADDRESS TABLE */
CREATE TABLE facility_address (
    facility_address_id INT NOT NULL AUTO_INCREMENT,
    facility_address varchar(255) NOT NULL,
    city varchar(50) NOT NULL,
    province varchar(50) NOT NULL,
    postal_code varchar(10) NOT NULL,

    PRIMARY KEY (facility_address_id)
);
```

The primary key for this table is ‘facility_address_id’ and it is of type INT. This key uniquely identifies the ID number of the facility address which is then stored into the ‘Facility’ table. The actual address of the facility is stored in the attribute ‘facility_address’ which is of type VARCHAR and has a length of 255. The city and province are stored in attributes ‘city’ and ‘province’ respectively, which are both of type VARCHAR with length 50. The postal code is of type VARCHAR with a length of 10. None of these values can be NULL as it is necessary that an address is composed of each of these individual parts.

Work Table

```
CREATE TABLE work (
    work_id INT NOT NULL AUTO_INCREMENT,
    employee_medicare_number INT NOT NULL,
    role_id INT NOT NULL,
    facility_id INT NOT NULL,
    start_date DATE NOT NULL,
    end_date Date,


    FOREIGN KEY (employee_medicare_number) REFERENCES employee(medicare_number),
    FOREIGN KEY (role_id) REFERENCES role(role_id),
    FOREIGN KEY (facility_id) REFERENCES facility(facility_id),
    PRIMARY KEY (work_id),
    CONSTRAINT assignment CHECK (start_date < end_date
        OR end_date IS NULL),
    CONSTRAINT timeslot UNIQUE (employee_medicare_number , start_date , end_date)
);
```

This table holds the work related information for an employee. Specifically, it holds the start date of when the employee started working at a facility, the end date if there is one, the role of the employee at the facility and what facility the employee worked at. All of these attributes are considered NOT NULL with the exception of ‘end_date’ as an employee might still be working at a facility at present time. The employee is distinguished by their medicare number, which is a foreign key and it relates to the employee’s medicare number in the ‘medicare_number’ attribute from the ‘employee’ table. The attribute ‘role_id’ is a foreign key that relates to the attribute ‘role_id’ of the ‘role’ table. This attribute holds the ID of the name of the role that the employee works as (doctor, nurse, etc). The ‘facility_id‘ attribute is a foreign key that relates to the ‘facility_id’ of the ‘facility’ table and it represents what facility the employee works at. There have been constraints that have been added to the work table. These constraints verify if an employee is currently employed and is working at a facility. It cannot be the case that an employee is working at 2 different facilities at the same time.

Role Table

```
/* ROLE TABLE */
CREATE TABLE role (
    role_id INT NOT NULL AUTO_INCREMENT,
    name varchar(255) NOT NULL,

    PRIMARY KEY (role_id)
);
```

The primary key of the role table is “role_id” of type INT and this uniquely identifies the type of the role of the employee. This ID is used as a foreign key in the ‘work’ table. The attribute ‘name’ holds the name of the role (doctor, nurse, etc) and is of type VARCHAR with a length of 255. Neither of these attributes can be NULL as there must be an ID number that distinguishes each role in the database and each of these roles have a name.

Patient Vaccine Table

```
/* VACCINE TABLE */
CREATE TABLE patient_vaccine (
    vaccine_id INT NOT NULL AUTO_INCREMENT,
    employee_id INT NOT NULL,
    vaccine_type_id INT NOT NULL,
    date DATE NOT NULL,
    location varchar(255) NOT NULL,
    dose_number INT,

    FOREIGN KEY (employee_id) REFERENCES employee(medicare_number),
    FOREIGN KEY (vaccine_type_id) REFERENCES vaccine_type(vaccine_type_id),
    PRIMARY KEY (vaccine_id)
);
```

This table references the vaccination information of an employee in the database. The primary key for the ‘patient_vaccine’ table is ‘vaccine_id’ and it is of type INT. This attribute uniquely identifies each time an employee is vaccinated, along with information pertaining to the employee, the type of vaccine, the date it was taken, where it was taken and how many doses.

The attribute ‘employee_id’ is a foreign key of type INT which relates to the ‘medicare_number’ of the ‘employee’ table as this medicare number is unique for all employees in the database. The ‘vaccine_type_id’ attribute is a foreign key of type INT which relates to the ‘vaccine_type_id’ attribute of the ‘vaccine_type’ table and this holds the information regarding what kind of vaccine was given to the employee (Pfizer, Moderna, etc). The ‘date’ attribute is of type DATE and it contains the date the employee was vaccinated. None of these attributes can be NULL as it is necessary that the database holds information about when the vaccine was done, where it was done, the type of vaccine and who it was given to. The dosage of vaccine can be NULL as it is possible an employee has not been vaccinated.

Vaccine Type Table

```
/* VACCINE TYPE */
CREATE TABLE vaccine_type (
    vaccine_type_id INT NOT NULL AUTO_INCREMENT,
    name varchar(255) NOT NULL,

    PRIMARY KEY (vaccine_type_id)
);
```

The primary key for this table is ‘vaccine_type_id’ is of type INT and it uniquely identifies what kind of vaccine the employee has taken. The ‘name’ attribute is a VARCHAR of length 255 and holds the name of the vaccine type (Pfizer, Moderna, etc). Neither of these values can be NULL as a vaccine must have a name and the database must have an ID to distinguish from among the different types of vaccines.

Contracted Table

```
/* CONTRACTED TABLE */
CREATE TABLE contracted (
    contracted_id INT NOT NULL AUTO_INCREMENT,
    employee_id INT NOT NULL,
    disease_id INT NOT NULL,
    date DATE NOT NULL,


    FOREIGN KEY (employee_id) REFERENCES employee(medicare_number),
    FOREIGN KEY (disease_id) REFERENCES disease(disease_id),
    PRIMARY KEY (contracted_id)
);
```

The primary key for the ‘contracted’ table is ‘contracted_id’ and it is of type INT. This key uniquely identifies the employees that have contracted a virus. The employee is referenced through a foreign key named ‘employee_id’ and this is linked to the medicare number of the employee, namely in the ‘medicare_number’ attribute of the ‘employee’ table. The diseases that the employee can contract each have an ID number related to them and they are stored in the ‘disease_id’ attribute, which is a foreign key of type INT and it is linked to the ‘disease_id’ attribute of the ‘disease’ table. The date that the employee contracted the disease is stored in the ‘date’ attribute and it is of type ‘DATE’. None of these attributes can be NULL as a disease that has been contracted by an employee must be contained within some form in the database, it must be linked to an employee, there must be a disease from which the employee had contracted and there must be a date they would have contracted it.

Disease Table

```
CREATE TABLE disease (
    disease_id INT NOT NULL AUTO_INCREMENT,
    name varchar(255) NOT NULL,

    PRIMARY KEY (disease_id)
);
```

The primary key for this table is ‘disease_id’ of type INT and it is used to uniquely identify the type of diseases the employees can contract. The name of the disease is stored in the attribute ‘name’ and it is of type VARCHAR with a length of 255. These cannot be NULL as the database needs an ID for each disease and they must have a name to distinguish them.

3. Queries

1.

```
SELECT facility_address.province, facility.name as "Facility Name", facility.capacity,
COUNT(*) as "Employees currently Working"

FROM facility, work, facility_address

WHERE work.end_date IS NULL AND facility.facility_id = work.facility_id AND
facility_address.facility_address_id = facility.facility_address_id

GROUP BY work.facility_id

ORDER BY name;
```

Result as Table:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

province	Facility Name	capacity	Employees currently Working
Province3	Bio Pharmacy	200	1
Province2	Green Clinic	1200	2
Province1	Hospital Maisonneuve Rosemont	1000	6
Province5	THE CLSC cote des neiges	50	2

This query selects data from the tables facility, work, and facility_address to retrieve information about the number of employees working in each facility, along with the facility's name, capacity, and province. The group by statement groups the result by work.facility_id so that the function calculates the number of employees working at each facility and orders by name in the ascending order.

2.

```
SELECT facility_address.province, facility_type.name AS "Facility Type", COUNT(*)
AS "Count of this Type"

FROM facility, facility_address, facility_type

WHERE facility_address.facility_address_id = facility.facility_address_id

AND facility_type.facility_type_id = facility.facility_type_id

GROUP BY province, facility.facility_type_id

ORDER BY province ASC, COUNT(*) DESC;
```

Result as table:

	province	Facility Type	Count of this Type
▶	Province1	Hospital	1
	Province1	Research Facility	1
	Province2	Clinic	1
	Province2	Distributor	1
	Province3	Pharmacy	1
	Province3	dentist	1
	Province4	Special Installment	1
	Province4	eye doctor	1
	Province5	CLSC	1
	Province5	psychiatrist	1

This query retrieves information of facilities by province, facility type and the number of facility type in each province. It joins three tables facility, facility_address, and facility_type where the ID matches to address, types, and facility table. It then groups by the result by province, Facility type and the number of each facility in that province. The results are then orders by ascending order of the province and descending order of type of each facility.

3.

```
SELECT emp.first_name, emp.last_name, rl.name AS
"role",emp.medicare_number, emp.phone_number, emp.email, ct.date,
ct.disease_id, ds.name

FROM employee emp

JOIN work wk ON emp.medicare_number = wk.employee_medicare_number

JOIN role rl ON wk.role_id = rl.role_id

LEFT JOIN contracted ct ON emp.medicare_number = ct.employee_id

LEFT JOIN disease ds ON ct.disease_id = ds.disease_id

WHERE wk.employee_medicare_number = emp.medicare_number AND
wk.facility_id = 1 AND (wk.role_id = 5 OR wk.role_id=6) AND ct.disease_id=1 AND
wk.end_date IS NULL

AND ct.employee_id = emp.medicare_number AND ds.name = "COVID-19"

ORDER BY ct.date DESC, emp.first_name ASC, emp.last_name ASC;
```

Result as table:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	role	medicare_number	phone_number	email	date	disease_id	name
emp	fourteen	nurse	14	1234543245	emp.fourteen@gmail.com	2021-01-12	1	COVID-19
emp	thirteen	nurse	13	1486539473	emp.thirteen@gmail.com	2020-07-01	1	COVID-19

This query selects date and diseas_id from contracted table, name, first_name, last_name, medicare_number, phone_number and email from employee table. These data are selected when these conditions are met listed down below

1) employee_medicare_number column in the work table matches the medicare_number column in the employee table.
2) facility_id column in the work table must equal 1,where 1 denotes the Hospital Maisonneuve Rosemont.

3) role_id column in the work table must be either 5 or 6, 5 denotes nurses and 6 denotes doctors.
4) disease_id column in the contracted table must equal 1, 1 denoted COVID-19
5) end_date column in the "work" table must be NULL, which means the employee is currently working.
6) employee_id column in the contracted table must match the medicare_number column in the employee table.
7) name column in the disease table must be COVID-1'.

4.

```
SELECT emp.first_name, emp.last_name, rl.name AS "role",
emp.medicare_number, emp.phone_number, emp.email, pv.vaccine_type_id,
ds.name

FROM employee emp

JOIN work wk ON emp.medicare_number = wk.employee_medicare_number

JOIN role rl ON wk.role_id = rl.role_id

LEFT JOIN patient_vaccine pv ON emp.medicare_number = pv.employee_id

LEFT JOIN contracted ct ON emp.medicare_number = ct.employee_id

left JOIN disease ds ON ct.disease_id = ds.disease_id

WHERE wk.facility_id = 1 AND pv.employee_id IS NULL AND (ct.disease_id IS NULL
OR ct.disease_id > 1)

ORDER BY rl.name ASC, emp.first_name ASC, emp.last_name ASC;
```

Result as table:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	role	medicare_number	phone_number	email	vaccine_type_id	name
emp	five	doctor	5	1567901234	emp.five@email.com	NULL	Beta Variant
emp	five	doctor	5	1567901234	emp.five@email.com	NULL	SARS-Cov-2
emp	five	doctor	5	1567901234	emp.five@email.com	NULL	SARS-Cov-2
emp	fifteen	Receptionist	15	1423786987	emp.fifteen@gmail.com	NULL	NULL

This query selects data from tables employee, work, role, patient_vaccine, contracted, and disease. It joins these tables using left join and inner join statements to gather employee

information,their first and last name, role, medicare number, phone number, and email. It includes information on whether the particular employer have a vaccine or contracted a disease, such as the vaccine type ID and disease name. These data are selected only for employees currently working in the Hospital Maisonneuve Rosemont. and have not received a vaccine, and have not contracted a specific disease. It then sorts the results by role name, first name, and last name in ascending order.

5.

```
SELECT emp.first_name, emp.last_name,wk.start_date, wk.end_date, fac.name as "facility Name", rl.name AS "role",emp.medicare_number, emp.phone_number, emp.email

FROM employee emp

JOIN work wk ON emp.medicare_number = wk.employee_medicare_number

JOIN role rl ON wk.role_id= rl.role_id

JOIN facility fac ON wk.facility_id = fac.facility_id

WHERE emp.medicare_number in (select employee_medicare_number from work group by employee_medicare_number Having count(employee_medicare_number)>1)

ORDER BY emp.first_name ASC, emp.last_name ASC, wk.start_date ASC;
```

Result as table:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	start_date	end_date	facility Name	role	medicare_number	phone_number	email
emp	fourteen	2002-01-01	2003-01-01	craigs special installment	security	14	1234543245	emp.fourteen@gmail.com
emp	fourteen	2008-09-20	NULL	Hospital Maisonneuve Rosemont	nurse	14	1234543245	emp.fourteen@gmail.com
emp	six	2021-01-01	2021-01-10	Green Clinic	Admin	6	1276367890	emp.six@email.com
emp	six	2022-01-01	2022-02-02	Hospital Maisonneuve Rosemont	nurse	6	1276367890	emp.six@email.com

This query retrieves employee information along with their work history and facility information for those employees who have worked in at least 2 facilities. The query joins the employee, work, role and facility tables using various foreign keys, and filters the results by selecting only those employees whose medicare_number is present in a subquery that returns the employee_medicare_number for all employees who have worked at more than one facility. Finally, the results are ordered alphabetically by employee name and chronologically by work start date.

6.

```
SELECT name, SUM(dose_number) AS Total_doses FROM vaccine_type
JOIN patient_vaccine ON vaccine_type.vaccine_type_id =
patient_vaccine.vaccine_type_id
GROUP BY vaccine_type.name
ORDER BY Total_doses desc;
```

Result as table:

name	Total_doses
Pfizer	7
Moderna	4
AstraZeneca	3

This query is used to get the total number of doses administered for each type of vaccine. It joins the vaccine_type table with the patient_vaccine table on the vaccine_type_id field. The SUM() function is used to calculate the total number of doses administered for each vaccine type. The result is grouped by the name of the vaccine type using the group by clause. Finally, the order by clause is used to sort the result set in descending order of the total number of doses administered.

7.

```
SELECT name AS disease_name, province, COUNT(employee_id) AS num_infected
FROM contracted JOIN disease ON contracted.disease_id = disease.disease_id
JOIN employee ON contracted.employee_id = employee.medicare_number
JOIN employee_address ON employee.employee_address_id =
employee_address.employee_address_id
GROUP BY disease.name , employee_address.province
ORDER BY disease.name ASC , num_infected DESC;
```

Result as table:

disease_name	province	num_infected
Alpha Variant	Province1	1
Beta Variant	Province3	1
Beta Variant	Province1	1
COVID-19	Province1	6
COVID-19	Province3	1
Delta Variant	Province1	1
SARS-Cov-2	Province3	2
SARS-Cov-2	Province1	1

This query is used to retrieve the number of employees who have contracted each disease in each province. The query joins four tables: contracted, disease, employee, and employee_address. It retrieves the name of the disease, the province of the employee, and the number of employees who have contracted the disease in that province. The COUNT function is used to count the number of employee IDs for each group of disease name and province. The group by clause groups the result by disease name and employee's province, so that the COUNT function can be applied to each group. The order by clause sorts the result in ascending order by disease name and descending order by the number of employees infected.

8.

```
SELECT emp.first_name, emp.last_name,  wk.start_date, fac.name as "facility Name", rl.name AS "role", emp.medicare_number, emp.phone_number, emp.email, count(*) as "total number of infections"

FROM employee emp

JOIN work wk ON emp.medicare_number = wk.employee_medicare_number

JOIN role rl ON wk.role_id= rl.role_id

JOIN facility fac ON wk.facility_id = fac.facility_id

JOIN contracted ct ON emp.medicare_number = ct.employee_id

WHERE wk.end_date IS NULL AND (emp.medicare_number in (Select employee_id from contracted group by employee_id having count(*)>=3))

GROUP BY  emp.first_name, emp.last_name, rl.name, emp.phone_number, emp.email, wk.start_date, wk.end_date, fac.name

order by count(*) DESC, emp.first_name DESC, emp.last_name DESC;
```

Result as table:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	start_date	facility Name	role	medicare_number	phone_number	email	total number of infections
emp	two	2022-01-01	Hospital Maisonneuve Rosemont	Receptionist	2	1347242901	emp.two@email.com	4
emp	five	2022-01-01	Hospital Maisonneuve Rosemont	doctor	5	1567901234	emp.five@email.com	3

This SQL query retrieves employee information who have been infected 3 times or more for each employee who is currently working at a facility and gets the total number of infections. It joins tables employee, work, role, facility, and contracted, to retrieve the necessary data. The query filters out employees who have already left their jobs and groups the result by various employee and job-related attributes. It then counts the total number of infections for each employee using the COUNT() function and displays the results in ascending order of the

employee's first name, last name, role, phone number, email, start date, end date, and facility name.

4. Populate tables

The following SQL insert statements were used to populate the tables and perform the queries. Only the first 10 values entered into each table are covered by the insert statements shown below.

```
/* ------------------------------------------------------------ */
/* enter roles */
INSERT INTO role (name)
VALUES   ('Admin'),  ('Receptionist'),  ('pharmacist'),  ('cashier'), ('doctor'), ('nurse'), ('security'),
('regular employee'),('HR manager'),('legal');

/* ------------------------------------------------------------ */
/* enter facilities type*/
INSERT INTO facility_type (name) VALUES ('Hospital'),('Clinic'),('Pharmacy'),('Special
Installment'),('CLSC'),('Research Facility'),('Distributor'),('dentist'),('eye doctor'),('psychiatrist');

/* ------------------------------------------------------------ */
/* enter diseases*/
INSERT INTO disease (name)VALUES
("COVID-19"),("SARS-Cov-2"),("Delta Variant"),("Alpha Variant"),("Beta Variant"), ("Common
cold"),("Chicken pox"),("Malaria"),("Migraine"),("UTI");

/* ------------------------------------------------------------ */
/* enter facilities address*/
INSERT INTO facility_address (facility_address, city, province, postal_code) VALUES ('123
Street', 'town1', 'Province1', 'ABC DEF'),
('456  Street', 'town2', 'Province2', 'GHI JKL'),
('789  Street', 'town3', 'Province3', 'MNO PQR'),
('101112  Street', 'town4', 'Province4', 'STU VWX'),
('131415  Street', 'town5', 'Province5', 'XYZ 123'),
('434  Street', 'town4', 'Province1', 'FWC DEF'),
('645  Street', 'town5', 'Province2', 'KWI FWL'),
('756  Street', 'town2', 'Province3', 'GNO QSR'),
('675  Street', 'town7', 'Province4', 'WEU FEX'),
('567  Street', 'town8', 'Province5', 'FEW FWE');

/* ------------------------------------------------------------ */
/* enter facilities */
INSERT INTO facility (facility_type_id, facility_address_id, name, phone_number, website_url,
capacity) VALUES (1, 1, 'Johns Hospital', 1234567890, 'www.johnhospital.com', 1000),
(2, 2, 'Green Clinic', 1234567890, 'www.greenclinic.com', 1200),
```

```
(3, 3, 'Bio Pharmacy', 1234567890, 'www.biophar.com', 200),
(4, 4, 'craigs special installment', 1234567890, 'www.craigsSI.com', 800),
(5, 5, 'THE CLSC', 1234567890, 'www.THEclsc.com', 50),
(6, 6, 'MARTINSK research', 1234567890, 'www.mskRESEARCH.com', 10000),
(7, 7, 'Damien & co distributor',          1432567890, 'www.DCdistributor.com', 1200),
(8, 8, 'fun dentist ',  1242567890, 'www.funDentist.com', 20),
(9, 9, ' eye GUY', 1654567890, 'www.EyeGUY.com', 8),
(10, 10, 'Psych help',1878567890, 'www.PHelp.com', 5);


/* ------------------------------------------------------------ */
/* enter Vaccines*/
INSERT INTO vaccine_type (name) VALUES ("Pfizer"),("Moderna"),("AstraZeneca"),("Johnson
& Johnson"),("Covifenz"),
("Rabies Vaccine"),("Chicken pox vaccine"),("flu shot"),("saline shot dummy vaccine"),("Tetanus
vaccine");

/* ------------------------------------------------------------ */
/* enter employee addresses*/
INSERT INTO employee_address (employee_address, city, province, postal_code) VALUES
('1 employee add', 'montreal', 'Province1', 'ABC DEF'),
('2 employee add', 'GTA', 'Province1', 'AQC DGS'),
('3 employee add', 'Phil', 'Province1', 'SDF ASD'),
('4 employee add', 'ottawa', 'Province4', 'ASD HJJ'),
('5 employee add', 'vancover', 'Province3', 'YHJ HFH'),
('6 employee add', 'montreal', 'Province1', 'ABC DEF'),
('7 employee add', 'GTA', 'Province1', 'AQC DGS'),
('8 employee add', 'Phil', 'Province1', 'SDF ASD'),
('9 employee add', 'ottawa', 'Province4', 'ASD HJJ'),
('10 employee add', 'vancover', 'Province3', 'YHJ HFH');



/* ------------------------------------------------------------ */
/* enter employees*/
INSERT INTO employee (employee_address_id, first_name, last_name, date_of_birth,
phone_number, email) VALUES
 (1, 'emp', 'one', '2000-01-01', 1234567890, 'emp.one@email.com'),
 (2, 'emp', 'two', '2001-05-03',  2347242901, 'emp.two@email.com'),
 (3, 'emp', 'three','2000-01-03', 5645689012, 'emp.three@email.com'),
 (4, 'emp', 'four', '2010-07-05', 4546590123, 'emp.four@email.com'),
 (5, 'emp', 'five', '2005-01-10', 7567901234, 'emp.five@email.com'),
 (6, 'emp','six', '2000-01-01', 1234567890, 'emp.six@email.com'),
 (7, 'emp', 'seven', '2001-05-03',  2347242901, 'emp.seven@email.com'),
```

```
 (8, 'emp', 'eight','2000-01-03', 5645689012, 'emp.eigh'@email.com'),
 (9, 'emp', 'nine', '2010-07-05', 4546590123, 'emp.nine@email.com'),
 (10, 'emp', 'ten', '2005-01-10', 7567901234, 'emp.ten@email.com');

/* ------------------------------------------------------------ */
/* enter employee - facility work relationship*/
INSERT INTO work (employee_medicare_number, role_id, facility_id, start_date, end_date)
VALUES
(1, 1, 1, '2020-01-01', '2022-01-10'),
 (2, 2, 1,  '2022-01-01', null),
 (3, 3, 1,  '2022-01-01', null),
(4, 4, 1,  '2022-01-01', '2022-06-01'),
 (5, 5, 1,  '2022-01-01', null),
(6, 1, 2, '202`-01-01', '202`-01-10'),
 (7, 2, 2,  '2020-01-01', null),
 (8, 3, 2,  '2020-01-01', null),
(9, 1, 3,  '2021-01-01', '2022-06-01'),
 (10, 2, 3,  '2018-01-01', null);


/* ------------------------------------------------------------ */
/* enter contracted diseases*/

INSERT INTO contracted (employee_id, disease_id, date) VALUES
(1,1,"2020-01-19"),
(2,3,"2020-01-20"),
(2,1,"2020-01-21"),
(2,2,"2020-01-21"),
(1,4,"2020-01-23"),
(5,2,"2020-01-24"),
(1,1,"2022-12-12"),
(1,1,"2022-12-22"),
(6,1,"2021-12-01"),
(5,2,"2022-01-10");


/* ------------------------------------------------------------ */
/* enter patient vaccine taken*/
INSERT INTO patient_vaccine (employee_id, vaccine_type_id, date, location,
dose_number)VALUES
(1,1,'2021-05-03','Montreal', 1),
(2,3,'2021-04-03','Vancover', 1),
(3,1,'2021-03-03','Montreal', 1),
(4,2,'2021-05-03','Toronto', 1),
```

```
(6,1,'2021-08-03','montreal', 1),
(7,2,'2021-01-03','montreal', 1),
(1,1,'2021-08-03', 'montreal',2),
(2,3,'2021-08-03', 'montreal',2),
(4,2,'2021-08-03','Toronto', 2),
(6,1,'2021-12-03','montreal', 2);
```

5. SELECT COUNT(*) output for each relation

The following are the screenshots for the SELECT COUNT(*) from each of the 11 tables. This displays the total rows/ tuples from each of the tables.

```
1    SELECT COUNT(*) FROM contracted
```

Result Grid | Filter Rows: | Expor

COUNT(*)
14

```
1    SELECT COUNT(*) FROM disease
```

Result Grid | Filter Rows:

COUNT(*)
10

```
1    SELECT COUNT(*) FROM employee
```

Result Grid | Filter Rows:

COUNT(*)
16

```
1    SELECT COUNT(*) FROM employee_address
```

Result Grid | Filter Rows: | Export:

	COUNT(*)
▶	16

```
1    SELECT COUNT(*) FROM facility
```

Result Grid | Filter Rows:

	COUNT(*)
▶	10

```
1    SELECT COUNT(*) FROM facility_address
```

Result Grid | Filter Rows: | Export:

	COUNT(*)
▶	10

```
1    SELECT COUNT(*) FROM facility_type
```

Result Grid | Filter Rows: | Expo

	COUNT(*)
▶	10

```
1    SELECT COUNT(*) FROM patient_vaccine
```

Result Grid | Filter Rows: | Expor

	COUNT(*)
▶	10

```
1    SELECT COUNT(*) FROM role
```

Result Grid | Filter Rows:

	COUNT(*)
▶	10

```
1    SELECT COUNT(*) FROM vaccine_type
```

Result Grid | Filter Rows:

	COUNT(*)
▶	10

```
1    SELECT COUNT(*) FROM work
```

Result Grid | Filter Rows:

	COUNT(*)
▶	18

Note : The table creation script, along with the query scripts and insertion data script are present on concordia's assigned project folder.
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